

BLIGH REEF LIGHT, AK, USA

WMO: 994680

Lat: 60.839N

Lon: 146.884W

Elev: 0

StdP: 14.70

Time Zone: -9.00 (NAK)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 18.5	(c) 20.6	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 46.2	(k) 27.1	(l) 42.6	(m) 26.2	(n) 32.2	(o) 20	(p) 0.773

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.5	(c) 64.1	(d) N/A	(e) 61.4	(f) N/A	(g) 59.6	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 9.5	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 40.5	(b) 35.1	(c) 31.4	(d) DB	(e) 15.7	(f) 69.3	(g) 4.4	(h) 4.2	(i) 12.5	(j) 72.3	(k) 9.9	(l) 74.7	(m) 7.5	(n) 77.0	(o) 4.2	(p) 80.0	(q) N/A
			(d) WB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.8	32.8	33.1	33.4	38.6	46.1	51.7	54.9	55.0	50.5	44.3	36.7	35.3		
	DBStd	9.47	6.10	4.93	4.82	3.66	4.10	3.49	3.06	3.14	3.06	3.72	6.10	5.39		
	HDD50	3077	533	472	515	343	135	17	1	0	30	177	399	455		
	HDD65	8118	998	892	980	793	587	399	313	310	436	641	849	920		
	CDD50	435	0	0	0	0	13	68	153	156	44	1	0	0		
	CDD65	1	0	0	0	0	0	0	0	1	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	12.7	18.3	16.7	17.2	9.3	7.9	6.8	7.0	8.2	11.0	13.1	19.1	18.4		
(15) Precipitation	PrecAvg	56.1	5.4	4.7	3.8	2.9	2.9	2.8	3.8	5.2	7.0	6.1	5.4	6.1		
	PrecMax	72.1	12.8	10.1	9.6	5.6	5.9	4.5	6.3	9.5	11.9	12.3	12.5	12.0		
	PrecMin	38.8	1.2	0.1	1.1	0.9	0.8	0.8	2.0	1.1	2.8	3.0	0.1	2.0		
	PrecStd	7.0	2.4	2.1	2.1	1.1	1.4	0.9	1.0	2.2	2.4	2.4	2.9	2.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.9	43.2	44.4	51.5	60.8	67.1	66.4	68.4	60.0	53.4	48.6	48.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	42.6	41.0	41.9	46.6	56.9	61.2	62.4	65.0	57.2	51.4	46.5	46.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	41.2	39.7	40.5	44.4	53.4	58.4	60.8	62.0	56.0	50.2	45.4	43.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	39.7	38.7	39.1	42.8	51.3	56.3	59.5	59.5	54.9	49.0	44.0	42.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.0	4.6	5.1	5.3	5.5	5.1	5.3	5.5	5.3	4.7	5.0	4.9		
		MCDBR	5.9	5.7	5.9	7.3	9.8	8.1	7.1	8.7	6.6	5.7	5.7	6.1		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.267	0.319	0.329	0.370	0.397	0.381	0.396	0.390	0.346	0.324	0.277	N/A		
	taud		2.072	2.132	2.173	2.119	2.149	2.307	2.309	2.316	2.406	2.271	2.061	N/A		
	Ebn at Noon		142	203	247	258	260	268	260	251	244	201	131	N/A		
	Edh at Noon		13	23	31	40	42	37	36	33	25	20	13	N/A		
(44) All-Sky Solar Radiation	RadAvg		83	284	745	1277	1640	1721	1509	1150	752	341	120	46		
	RadStd		11	51	96	125	152	116	131	142	120	43	25	5		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)		N/A	+2.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page